Ex.No.-3.1-Create 3D model from orthographic projection.

AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

1. Autodesk fusion 360.
2. Windows 10.
3. 16 GB of RAM (integrated graphics recommend 1 GB or more).
4. 2.5 Mbps or faster download; 500 Kbps or faster upload.

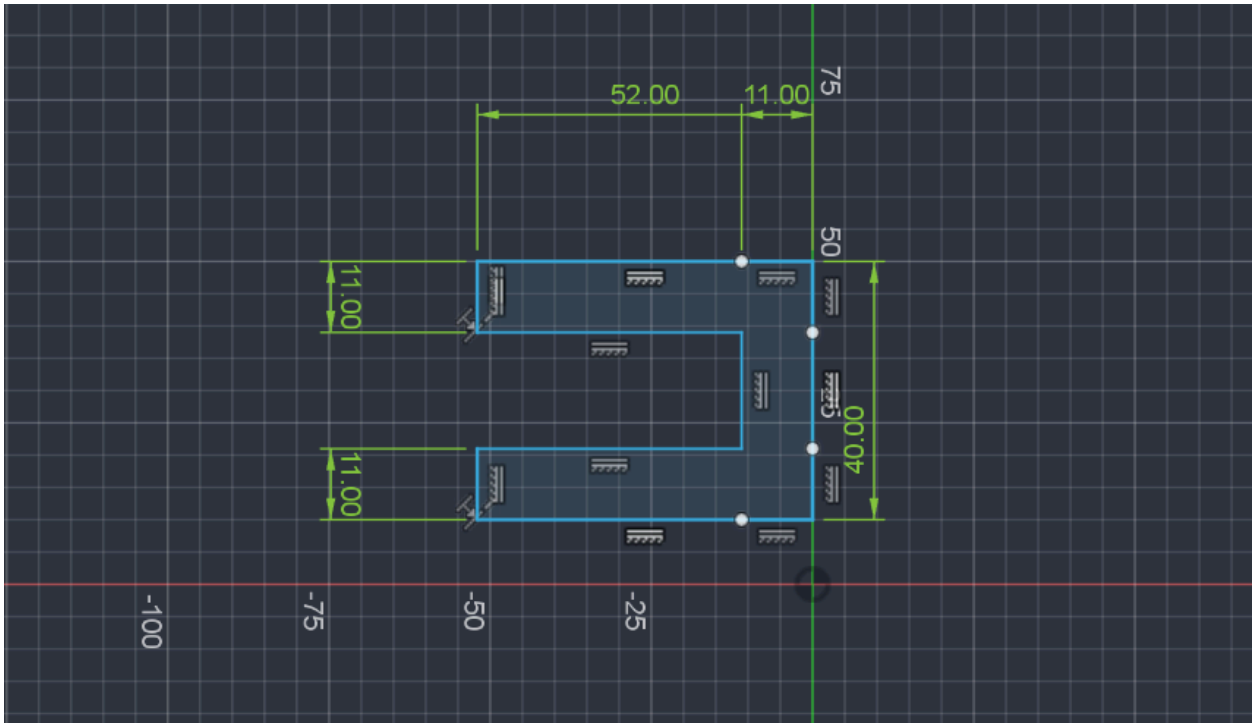
PROCEDURE:

STEP-1:
Open new file and then select sketch icon.
Select the TOP (XY) Plane

STEP-2:
Create a Sketch
Select Sketch > Rectangle > Center Rectangle.
Click once to begin placing the rectangle in the top left touching the y axis.
Type the Tab button and enter the values 52 mm and 40mm for the width and height respectively.

STEP-3:
Draw three more rectangles in three corners of the rectangle.
Let the dimensions be (52x11)mm, (52x11)mm, (40x11)mm.

Trim the excess.



Click Finish Sketch after completing.

STEP-4:
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of -32mm.
Press ENTER to accept the value.

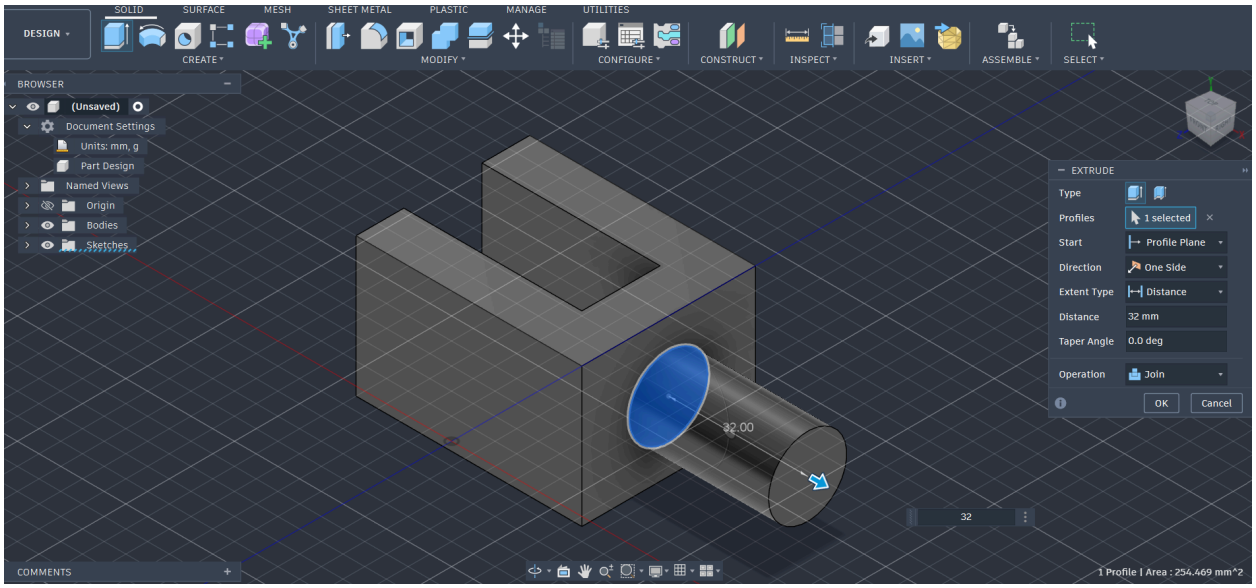
STEP-6:
Start the sketch command
Select Sketch > Create Sketch.

STEP-7:
Select the Sketch Plane
Select the right side of the 3D block

STEP-8:
Draw a circle of diameter 18mm at the centre.
Click Finish Sketch after completing.

STEP-9:
Select Extrude.
Select the sketch.

Click again to place the value.
Type in the value of 32mm.



Press ENTER to accept the value.

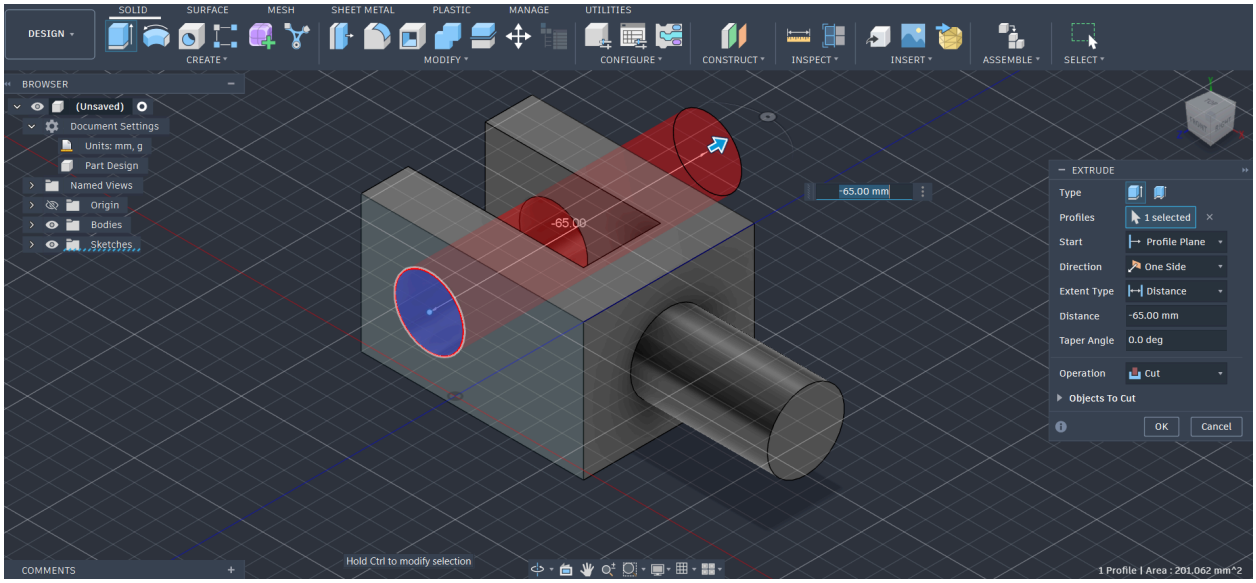
STEP-10:
Start the sketch command
Select Sketch > Create Sketch.

STEP-12:
Select the Sketch Plane
Select the top view of the 3D block

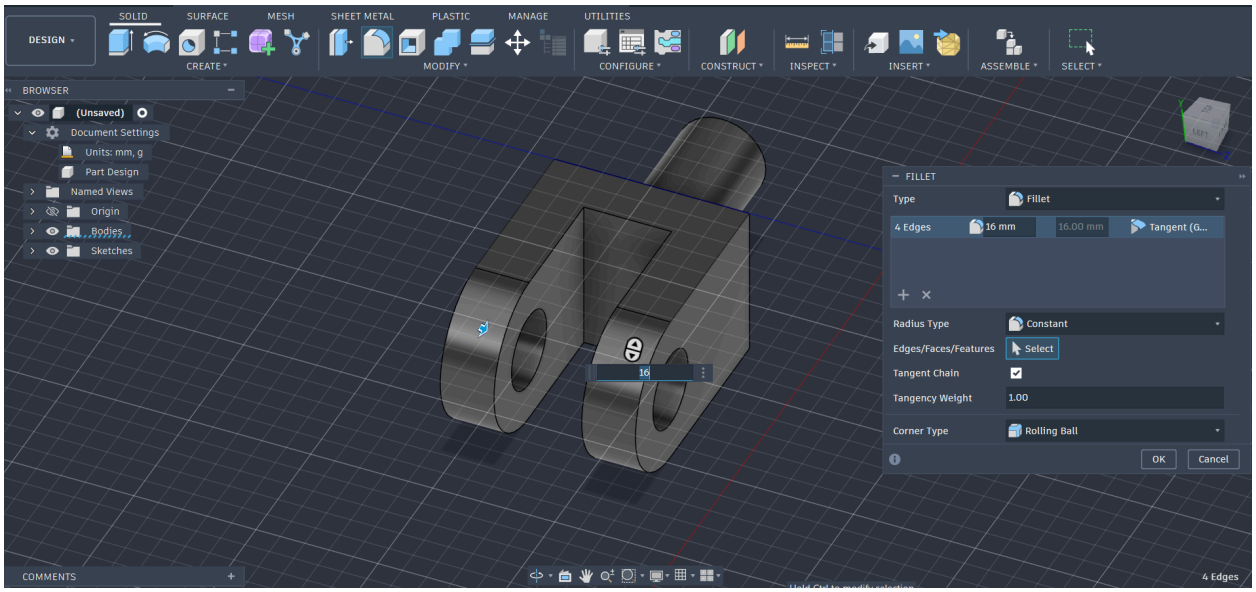
STEP-13:
Draw a circle of diameter 16mm.
Let the distance of the circle from the left corner of the rectangle be 36mm.
Let the distance of the circle from the top of the rectangle be 16mm.
Click Finish Sketch after completing.

STEP-14:
Select Extrude.
Select the sketch.
Choose the cut operation.

Drag the arrow till the portion is cut out.



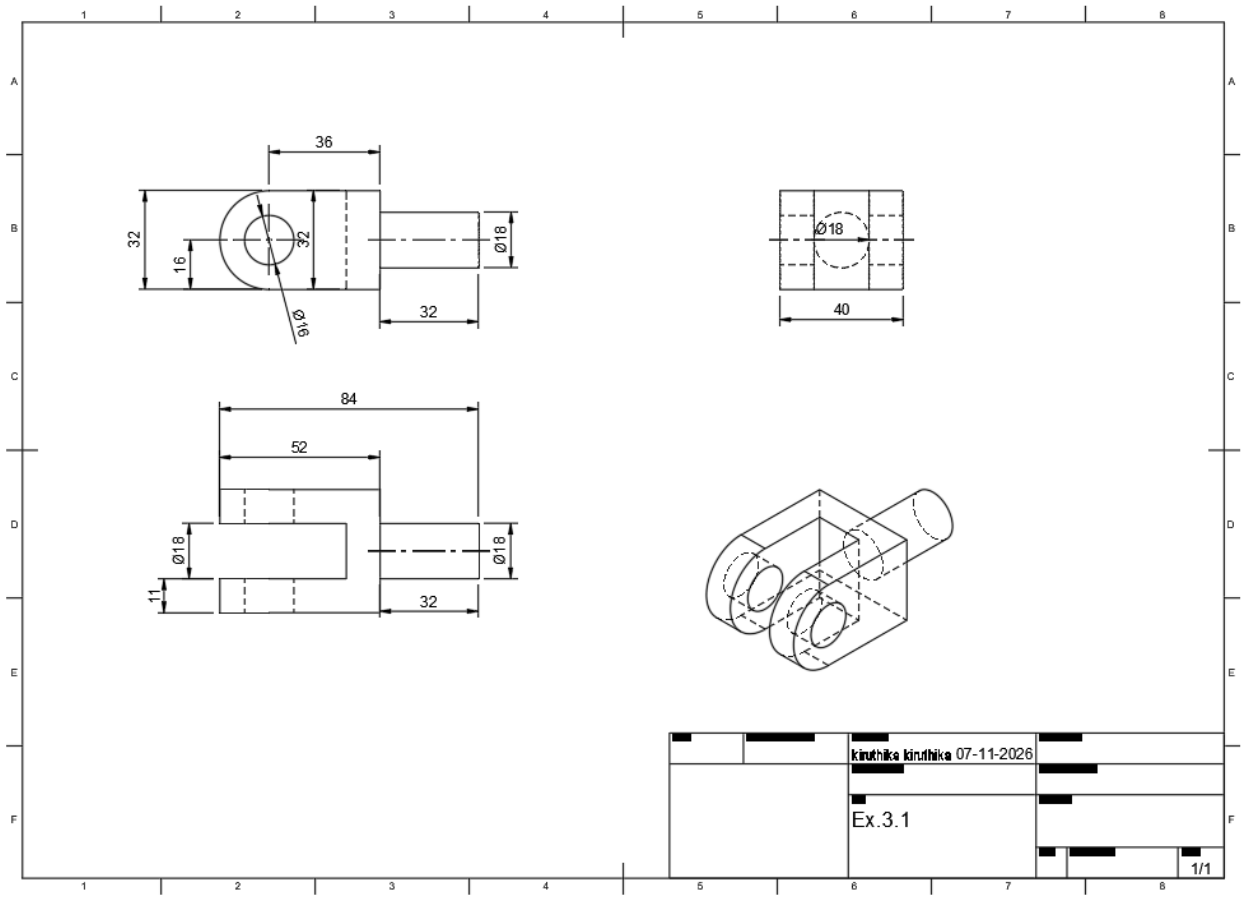
STEP-15:
Select Fillet.
Select the four corners of the rectangle.
Click again to place the value.
Type in the value of 16mm.



Press ENTER to accept the value.

STEP-16:
Switch to drawing mode to plot the sketch in ISO standard.

OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projection.